

Computer Vision Notes

1 Introduction

Computer Vision is a field of Artificial Intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos, and other visual inputs and take actions or make recommendations based on that information.

It's the science of teaching machines to "see" the world, interpret it, and make decisions based on what they observe.

1.1 What is Computer Vision?

Computer Vision enables systems to perform visual perception tasks such as identifying objects, tracking movement, recognizing patterns, and making sense of visual scenes automatically.

The following four areas are closely related but differ based on what kind of input they take and what kind of output they produce. Table 1 shows the relationship among the areas.

1.1.1 Image Processing

Image Processing focuses on improving, transforming, or analyzing digital images to enhance their quality or extract certain features. It deals with operations like noise removal, contrast enhancement, resizing, and edge detection. The goal is not to interpret what the image means, but to prepare or modify it for further analysis. For instance, before detecting a tumor in a medical scan, image processing might be used to sharpen or denoise the image for clearer visibility.

1.1.2 Computer Graphics

Computer Graphics is the field concerned with generating visual content from models, data, or textual descriptions. It focuses on creating or rendering images, animations, and 3D scenes that visually represent information. Applications include video games, simulations, movies, and virtual reality. While computer vision tries to understand the world from images, computer graphics tries to create a visual world from data.

1.1.3 Artificial Intelligence (AI)

Artificial Intelligence (AI) is a broader field that enables machines to mimic human intelligence, i.e., reasoning, learning, and decision-making. AI systems process symbolic or textual descriptions, recognize patterns, and make predictions or recommendations. When combined with vision or graphics, AI allows systems to think about what they see or generate realistic scenes intelligently. Examples include chatbots, autonomous decision systems, and intelligent assistants that adapt through experience.

Table 1: Relationship of Image Processing, Computer Vision, Computer Graphics, and AI

Input ↓ / Output →	Image Output	Description Output
Image Input	Image Processing Enhancing or transforming images (e.g., noise removal, edge detection, contrast adjustment).	Computer Vision Understanding images to extract meaning (e.g., object detection, facial recognition).
Description Input	Computer Graphics Generating visual content from models or data (e.g., 3D rendering, animation).	Artificial Intelligence (AI) Reasoning or decision-making based on symbolic or textual information (e.g., chatbots, expert systems).

1.1.4 Overlapping Applications

In practice, these fields often overlap. For example:

- **Autonomous Vehicles:** Combine *Computer Vision* (for perception), *Image Processing* (for filtering camera input), and *AI* (for decision-making).
- **Medical Imaging:** Uses *Image Processing* to enhance MRI or CT scans and *Computer Vision* to detect diseases.

- **Augmented Reality (AR):** Merges *Computer Graphics* (to render visuals) with *Computer Vision* (to understand the environment).
- **Robotics:** Integrates all four. Robots *see* using Computer Vision, *interpret* using AI, *enhance images* via Image Processing, and *display or simulate* environments using Computer Graphics.

2 Applications of Computer Vision

2.1 Object Detection and Recognition

Object detection identifies and locates specific objects within an image or video stream. It allows machines to answer “what” and “where” questions about visual scenes. Deep learning models like YOLO and Faster R-CNN are often used to detect faces, vehicles, animals, or tools. For example, in autonomous vehicles, computer vision systems detect pedestrians, traffic signs, and road lanes in real time to make driving decisions safely.

2.2 Facial Recognition

Facial recognition systems analyze facial features to identify or verify a person’s identity. They use patterns in the geometry of a face, such as distances between eyes, nose, and mouth, to create a unique representation. This technology is used in smartphone authentication, surveillance, and even attendance systems. For instance, unlocking an iPhone with Face ID is a direct example of computer vision performing facial recognition.

2.3 Medical Image Analysis

Computer vision plays a major role in medical diagnostics by analyzing X-rays, MRIs, and CT scans. It helps detect anomalies such as tumors, fractures, or infections automatically. Vision-based systems can highlight suspicious areas for doctors to review, improving diagnostic accuracy and reducing human error. For example, AI-based chest X-ray systems can identify early signs of pneumonia or COVID-19 faster than manual examination.

2.4 Autonomous Vehicles

Self-driving cars rely heavily on computer vision to perceive their surroundings. Cameras mounted on vehicles continuously capture the environment, while vision algorithms detect obstacles, lane markings, traffic signals, and other vehicles. For instance, Tesla’s Autopilot and Waymo’s vehicles use computer vision as a core component to navigate roads and make driving decisions in real time.

2.5 Industrial Inspection

In manufacturing, computer vision systems perform automatic visual inspection to detect defects, misalignments, or missing components on production lines. Unlike human inspectors, these systems operate continuously and with high precision. For example, in semiconductor manufacturing, vision-based inspection tools check microchips for surface defects that are invisible to the naked eye.

2.6 Gesture and Motion Recognition

Gesture recognition interprets human body movements or hand signs captured by a camera, enabling intuitive interaction with machines. Combined with motion tracking, it allows systems to understand physical actions. For instance, Microsoft's Kinect uses computer vision to detect body movements, letting users control games through gestures without a physical controller.

2.7 Optical Character Recognition (OCR)

OCR converts printed or handwritten text in images into editable digital text. It enables machines to read documents, signs, or receipts directly from images. For example, Google Lens can scan a photographed book page and convert it into searchable, selectable text. OCR is widely used in digitizing archives, automating form entry, and translating signs in real time.

2.8 Surveillance and Security Monitoring

Modern security systems use computer vision for real-time surveillance, detecting unusual behavior or unauthorized access. Vision algorithms analyze footage to identify intrusions, track people, or detect abandoned objects. For example, smart CCTV cameras in airports can alert authorities if someone enters a restricted zone or leaves a suspicious package unattended.

2.9 Agriculture and Environmental Monitoring

Computer vision helps monitor crop health, detect weeds, and estimate yields through drone or satellite imagery. It can also analyze environmental data like deforestation or pollution spread. For example, drones equipped with vision systems can detect pest infestations in cotton fields, allowing farmers to take early action and improve yield prediction.

2.10 Augmented Reality (AR) and Mixed Reality

Computer vision allows digital content to interact with the physical world in AR systems. It tracks surfaces, positions, and lighting to overlay 3D graphics accurately on real scenes. For example, in Pokémon Go, computer vision enables

the phone camera to detect the environment so that virtual characters appear naturally in the user’s surroundings.

3 Image Formation and Image Processing

3.1 Image Formation

Image formation is the process through which a camera captures an image by gathering light from an object. It typically involves three steps:

1. **Illumination:** Light from a source (e.g., sunlight or a bulb) falls on the object.
2. **Reflection:** Light interacts with the object, reflecting differently depending on its surface and color.
3. **Projection:** The reflected light passes through a lens and is focused onto the sensor, forming an image.

(a) 5x5 Intensity Matrix (Pixel Values 0–255)

0	50	100	150	200
20	70	120	170	220
40	90	140	190	240
60	110	160	210	255
80	130	180	230	255

(b) Grayscale Image Generated from Matrix

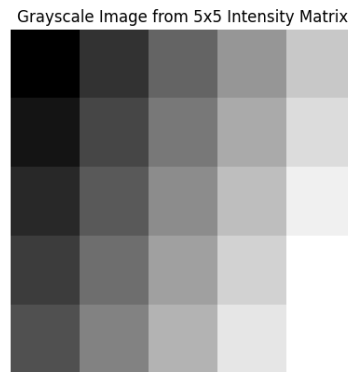


Figure 1: Relationship between Intensity Values and Grayscale Image Brightness

3.2 Image Representation

Every digital image is composed of tiny elements called **pixels**. Each pixel represents the smallest unit of an image and holds a numerical value known as its **intensity**. In a grayscale image, this intensity typically ranges from **0** to **255**, where **0** represents black, **255** represents white, and the values in between correspond to varying shades of gray. Collectively, these pixel intensity values form a two-dimensional matrix that defines the overall appearance of the image. The brightness of each pixel is directly proportional to its intensity value. The higher numbers mean lighter shades, while the lower numbers mean darker ones. Figure 1 (a) shows a small 5×5 matrix of pixel intensity values alongside (b) displays its corresponding grayscale image.

Interpretation: Each cell in the matrix represents a pixel intensity ranging from 0 (black) to 255 (white). As the intensity increases across rows and columns, the grayscale image becomes progressively brighter, demonstrating how pixel values form an image.

3.3 Image Processing

Image Processing involves manipulating or analyzing images to extract information or improve quality. Common techniques include:

- **Filtering:** Removes noise or emphasizes features (e.g., Gaussian blur, edge detection).
- **Transformation:** Alters appearance (rotation, scaling, translation).
- **Thresholding:** Converts an image into binary (black and white) to separate objects from the background.
- **Histogram Equalization:** Enhances contrast by spreading out intensity values for clearer details.

4 Pixel-Wise Operations in Images

In digital image processing, **pixel-wise operations** (also called **point operations**) are transformations applied individually to each pixel in an image, without considering its neighbors. These operations modify the intensity or color value of every pixel according to a defined rule or mathematical function.

Mathematically, if $I(x, y)$ represents the original image intensity at pixel position (x, y) , then a pixel-wise operation can be expressed as:

$$I'(x, y) = f(I(x, y))$$

where f is a transformation function that maps the input intensity to a new intensity.

Common Pixel-Wise Operations

- **Thresholding (Binarization):** Converts a grayscale image to binary by setting all pixels above a certain intensity to white and others to black.

$$I'(x, y) = \begin{cases} 255, & \text{if } I(x, y) > T \\ 0, & \text{otherwise} \end{cases}$$

Example: Used to separate an object from the background by choosing a threshold T (e.g., 128).

- **Brightness Adjustment:** Adds or subtracts a constant value β to make the image lighter or darker.

$$I'(x, y) = I(x, y) + \beta$$

Example: Increasing brightness by $\beta = 50$ makes all pixels lighter, provided the result stays within 0–255.

- **Contrast Stretching:** Expands the range of intensity values to enhance visibility of details. Contrast stretching, also known as **normalization**.

$$g(x, y) = \frac{(f(x, y) - f_{\min})}{(f_{\max} - f_{\min})} \times (L - 1)$$

Interpretation

Pixel-wise operations form the foundation for many higher-level image processing techniques. They are simple yet powerful tools to modify image tone, visibility, or prepare data for further analysis (e.g., edge detection or segmentation). Since each pixel is processed independently, these operations are computationally efficient and easy to parallelize.